

More examples of the method of indicators

1. Imagine the chance experiment of drawing a hand of five cards from a standard deck. Let N be the random variable that counts the number of suits in your hand. This variable takes the values 1, 2, 3, 4. For example, if you have 2 hearts, 1 diamond, and 2 clubs in your hand, then N for this outcome is 3.
 - a. Number the suits 1, 2, 3, 4 (in any way). Let A_i be the event that your hand has at least one card of suit i . Calculate $P(A_i)$.

$$P(A_i) = \frac{1}{2}$$

- b. Use the indicator variables for the events A_i to calculate $E[N]$.

2. A typical incoming first-year class at Hood has around 300 students. Imagine the chance experiment of gathering a first-year class. We will think about birthdays of students in the class, making two simplifying assumptions:
 - There are 365 possible birthdays; we ignore February 29th.
 - Birthdays for different students are independent.
 - a. What is the probability that some student in the 300-member incoming class has your birthday?

- b. What is the expected number of different birthdays in the 300-member incoming class?